Ship detection in Sentinel-1 radar against AIS: five pre-registered tests of what transfers to new water

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Abstract

A dark-vessel pipeline finds ships in Sentinel-1 radar and flags those that AIS does not explain. Tuned on one set of harbours, such a pipeline is usually reported on the same data. Here every choice that mattered was tested instead on new water, under rules committed before any of that water's data was fetched, and each test was scored once. Of five tests in 23 new US areas, plus a check on the Grand Banks of Newfoundland, three ideas did not transfer and one did. A learned detector that passed its promotion gate produced more false alarms than classical CFAR re-thresholded to find as many ships. A threshold set for a false-alarm budget gave 22.8 instead of 10 per 1,000 km^2. A map of recurring unexplained returns, built from each region's own first six months, cut dark-list false alarms by 34% in five new areas and by 65% in five structure-rich ones, once spots where vessels moor were left unmasked. A flag for vessels hiding beside structures caught about two thirds of injected vessels but raised too many false alarms in dense platform fields; calibrated on each area's own history, it missed too many vessels and still raised too many false alarms in five dense fields (9% caught, 1.04 added). Independent code review found twelve defects before the fourth test was frozen, and one published figure was corrected after the fact. All numbers are produced by scripts from public data.

1 Pipeline and data

Sentinel-1 IW ground-range scenes, radiometrically terrain-corrected to γ^0 at 10 m (Microsoft Planetary Computer; Torres et al. 2012; Small 2011), are read for each area's box. Open water comes from Overture Maps water polygons minus piers and bridges, shrunk 150 m off the coast. Detection is two-parameter CFAR on the cross-polarised band in dB (guard window 310 m, background 810 m, threshold k on the local z-score, fragments within 100 m merged, at least three pixels; Crisp 2004); production runs at $k = 5$. Truth is AIS from NOAA MarineCadastre for 2024 (Martin et al. 2025, CC0), interpolated to the overpass. A detection matches a ship inside a region shaped by the radar geometry: 300 m around the AIS position, stretched along the satellite track by the expected azimuth shift of a moving ship and along the course for a held fix. Detections within 150 m of a standing BOEM platform are left out of the score, as are AIS ships there. The headline metric is the probability of detection (Pd) at a fixed false-alarm rate, 5 per 1,000 km^2, read from the operating curve swept over k; false alarms are unmatched detections, an upper bound because vessels without AIS count against the detector.

2 Protocol

For each test, a script chose the passes by a fixed rule from scene metadata alone (for each area and month of 2024, the first scene covering at least 95% of the box), and wrote the rule, the areas, the endpoints and the reading rule to a configuration file, with SHA-256 hashes of the detector configuration and of every module the scorer uses. That file was committed before any radar or AIS data for those areas was fetched and pushed to GitHub before scoring (for the fourth and fifth tests, before fetching). The scorer refuses to run if a hashed file changed or if a result file already exists, so each test is scored once (Nosek et al. 2018). Table 1 lists the commits. Intervals are bootstrap percentiles (Efron and Tibshirani 1993; 20,000 draws). The pre-registered intervals of the first three tests resample passes within each area and hold the areas fixed; the fourth and fifth resample areas, then passes within each drawn area (Field and Welsh 2007), which is the interval for a new set of areas. Zero events are reported with the exact one-sided bound (Clopper and Pearson 1934).

Test	Rules committed	Scored	Pre-registered reading
1. Learned detector v6 vs CFAR	`033e970` 4 Nov 01:15	`8500765` 4 Nov 02:16	v6 worse on fresh water
2. False-alarm budget	`fdc8a64` 4 Nov 03:18	`58cfe24` 4 Nov 04:46	budget does not hold
3. Structure map	`24e4baf` 4 Nov 09:24	`13a0041` 4 Nov 10:20	improvement
Grand Banks installations	`d756a57` 4 Nov 18:11	`06c75db` 4 Nov 18:58	3 of 4 found
4. Map, occupied spots dropped; alongside flag	`82f9dbb` 5 Nov 00:21	`e0a83ab` 5 Nov 01:51	map improvement; flag does not transfer
5. Calibrated alongside flag in dense fields	`27dc282` 5 Nov 03:57	`de50f3b` 5 Nov 05:03	flag does not work; map no demonstrated improvement

Table 1: Pre-registrations: commit and time (UTC, 2026) from the repository history. The Grand Banks rules were a local commit; the others were pushed to GitHub before scoring, the fourth and fifth before their data was fetched.

3 Results

3.1 Test 1: a learned detector against CFAR

A Sentinel-1-pretrained ResNet-50 scoring CFAR's candidates (registry v6) passed the promotion gate on the development test passes. On 36 fresh passes in three new areas, its Pd at 5 FA minus CFAR's was -0.019 (-0.053 to $+0.021$), and at equal recall it produced 59% more false alarms than CFAR re-thresholded to $k = 6.5$ (15% to 116%). It was not promoted, and the gate gained an equal-recall rule that would have rejected it.

3.2 Test 2: a false-alarm budget

A threshold fixed at $k = 6$ to give 10 false alarms per 1,000 km^2 on 129 earlier passes gave 22.8 (21.3–24.4) on 60 passes in five new areas, and 58.3 in Mobile Bay. An adaptive threshold did not even out the areas and cost Pd (-0.047 at the same pooled false-alarm rate). Scored again with the platform exclusion, the pooled rate is 20.6; resampling the areas, its interval is 6.7–39.4 and includes the budget, and without Mobile Bay it is 10.1. The budget fails where structures are dense, not everywhere.

3.3 Test 3: a structure map from a region's own history

Most of Mobile Bay's excess sat on returns that recurred from pass to pass. The map adds no detector: a spot goes on it when returns that no AIS ship explains recur within 100 m on at least three of a region's January–June passes; on July–December passes, unexplained returns within 150 m of a spot leave the dark list, and returns an AIS ship explains are kept. In five new areas, false alarms at $k = 5$ fell 41% and Pd at 5 FA rose $+0.082$ as scored. *Correction:* the scorer had left out the platform exclusion; 8 BOEM platforms stand in the Sabine Pass box. Scored again with it (post hoc, after reproducing the published numbers exactly), the cut is 34% and the Pd gain $+0.046$, $+0.027$ to $+0.094$ with areas fixed and $+0.006$ to $+0.120$ with areas resampled. No AIS ship sat at a spot: 0 of 217 vessels, at most 1.4%.

3.4 The Grand Banks: does the map find the regulator's installations?

With no AIS and no list of structures, the map built from six HH/HV passes over the Jeanne d'Arc Basin found 3 of the 4 production installations on the C-NLOPB layer (Hibernia 85 m, Hebron 57 m, Terra Nova 134 m) and the newly installed West White Rose platform, with no other spot. SeaRose was missed: its return appears on 12 of 12 passes but 1,999 m from the listed point and within about 200 m of two independent published positions, and the turret-moored FPSO swings up to 382 m between passes, more than the map's 100 m rule.

3.5 Development: vessels at map spots, and a flag for vessels alongside

Masking structures has a cost. In Mobile Bay, 7 of 37 AIS ships on the July–December passes lay within 150 m of a spot built from January–June: some recurring returns are moorings and waiting berths. Leaving unmasked any spot that had an AIS ship beside it on a map pass cut that to 2, at the price of Mobile Bay's dark-list cut falling from 80% to 54%. A second idea flags a structure instead of hiding it: each structure's signature (separate

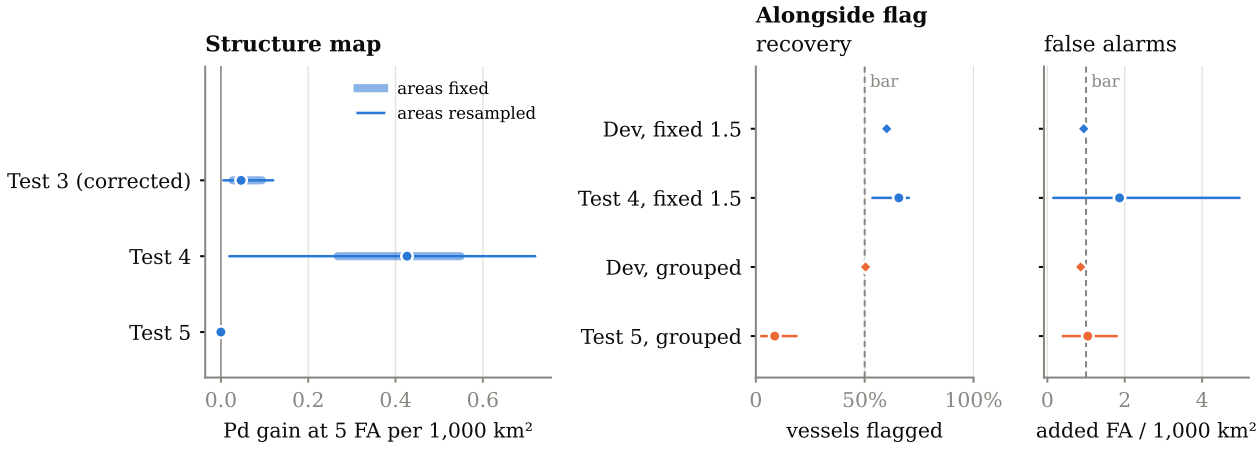


Figure 1: Left: the structure map's Pd gain at 5 false alarms per 1,000 km^2 (thick, areas fixed; thin, areas resampled). Middle and right: the alongside flag's recovery of injected vessels and its added false alarms, development (diamonds, no interval) and test (dots, areas resampled); dashed lines are the pre-registered bars.

returns within 150 m and the size of the largest) comes from the map passes, and a later pass flags it when it shows an extra return or one more than GROW times the largest seen. It was scored by adding real ship returns 60–140 m from structures (1,537 injections) and counting false flags as added false alarms. GROW = 1.5 was the most sensitive setting within 1.0 added false alarm per 1,000 km^2 (0.94): it flagged 60% of injected vessels (31%, 66% and 84% of small, medium and large returns). Three passes of independent review (Codex) found twelve defects in this scorer before it was frozen, among them a window that could cut a target and returns assigned to the wrong structure; fixing them lowered the recovery from 73% to 60%.

3.6 Test 4: structure-rich new water

Five areas chosen for their structures (four Louisiana fields with 56 standing platforms, and the Santa Barbara Channel, whose platforms are not on the federal Gulf list). The map, occupied spots dropped, cut false alarms by 65% (35–75%) and raised Pd at 5 FA from 0.50 to 0.93 (+0.019 to +0.720, areas resampled), with 2 of 115 AIS vessels at a spot; keeping the occupied spots would have exposed 7: an *improvement*. The alongside flag caught 66% of 1,547 injected vessels (54–70%), meeting its bar, but added 1.87 false alarms per 1,000 km^2 against a bar of 1.0: it *does not transfer*. Afterwards, 16 of the 23 false flags were found to come from Main Pass's 26-platform field; without it the rate would have been 0.67.

3.7 Test 5: a calibrated flag in dense platform fields

Two calibrations were developed on the ten areas used so far. One GROW per area, chosen by leave-one-pass-out on the area's own map passes, met the false-alarm bar (0.65) by switching the flag off in Main Pass, and caught only 42%. Calibrating structures with a neighbour within 500 m apart from isolated ones met both bars, narrowly (0.86 and 50%); 1,000 and 2,000 m did not. The fifth test applies it to the five densest federal fields in a scan of the Gulf (18 to 88 standing structures each). On 30 test passes (28 AIS ships) the calibrated flag caught 9% of 3,666 injected vessels (2–19%) for 1.04 added false alarms per 1,000 km^2 (0.40–1.79); 3 of 19 AIS ships beside a structure were flagged when scored as dark. For comparison, the fixed GROW = 1.5 caught 32% for 2.65. In fields this dense, even the fixed setting caught few: clustered platforms change too much from pass to pass for a changed-signature rule. The map, scored again, found 2 spots in all five fields and cut false alarms by 9% (0–21%), with 0 of 27 AIS vessels at a spot: almost every structure there is on the federal list that production already leaves out, so the map had little to add. The pre-registered readings: the calibrated flag *does not work*, and the map showed *no demonstrated improvement*.

4 What transferred, and what did not

- **Local evidence transferred; global settings did not.** A threshold and a learned detector tuned elsewhere failed on new water. A map built from each region's own history was an improvement in 2 of the 3 tests that scored it (in the fifth, almost every structure was already on the federal list, so it had little to add: its value is where structures are not listed), and its one transferable rule (leave a spot unmasked if an AIS ship sat there) was learned from a single bay and held on new water.
- **Dense structures are the hard case.** Mobile Bay decided the budget test, and Main Pass decided the flag test; in five dense fields even a calibrated flag caught only 9%. Pooled results hide this; leave-one-area-out numbers and area-resampled intervals show it.
- **AIS is limited truth near structures.** Many of the flag's false alarms in dense fields are probably small service boats without AIS: real vessels the metric counts against the detector. Progress there needs another kind of truth, such as analyst review.
- **Review and correction are part of the result.** Code review moved one development number by 13 points before a test, and one published figure (a 41% false-alarm cut, 34% once the platform list is applied) was corrected after the fact, with both versions kept.

5 Limitations

AIS-scored passes are US waters in 2024, VV/VH; the Grand Banks check is HH/HV with no AIS. Boats under 25 m are often missed at 10 m. Five areas per test give coarse area-resampled intervals. The structure map needs six months of a region's passes, and the alongside flag is scored on injected returns, which stand in for real vessels moored alongside.

6 Reproducibility

Every number here is read from a results file written by a script, and the note itself is built by `tools/build_note.py`. Data: contains modified Copernicus Sentinel data (2024–2026), via Microsoft Planetary Computer; AIS: NOAA Office for Coastal Management, CC0; water and piers: Overture Maps (includes OpenStreetMap data, ODbL); platforms: BOEM (U.S. Government), C-NLOPB, Government of Newfoundland and Labrador.

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